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We Measure What Changes

A proportionate evidence standard for learning and skills programmes delivered by small providers in low-resource settings

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HIGHLIGHTS

- Most evidence on education technology is thin or vendor-funded; only 7% of UK education technology companies had run randomised trials.
- A four-rung evidence ladder links the four levels of training evaluation to the tiers of evidence used in United States education law.
- Experimental studies from Kenya, Sub-Saharan Africa and India consistently favour instruction matched to learners' levels and sustained support for teachers.
- Effect sizes are context-dependent; local evidence should update, not be replaced by, results from elsewhere.
- A twelve-item minimum reporting standard and six testable propositions are offered for small providers and their funders.

ABSTRACT

Purpose: Funders, governments and employers increasingly ask training providers to show that their programmes work, yet the evidence base in education technology is thin and often supplied by vendors, and most workforce training evaluation stops at participant satisfaction. This paper proposes a proportionate evidence standard that a

small provider in a low-resource setting can meet. Design: A theory-building narrative review integrates the four levels of training evaluation with the evidence tiers used in United States education law and tests the resulting framework against experimental and quasi-experimental evidence from Kenya, Sub-Saharan Africa and other low- and middle-income countries. Findings: Experimental evidence consistently favours instruction matched to the learner's level and sustained, in-classroom support for teachers; effects of technology depend on the pedagogy it carries; teacher knowledge is a binding constraint in the region; and effect sizes vary with context and typically shrink at scale. The reviewed literatures on training transfer and publication bias show that reaction data are weak evidence of impact and that unreported null results distort the evidence base. Implications: Small providers can meet credible standards without large budgets by publishing a logic model, pre-specifying outcomes, using comparison groups where feasible, engaging an independent evaluator, disaggregating results, reporting costs, and publishing null findings. A four-rung evidence ladder, a twelve-item minimum reporting standard and six testable propositions are offered. Originality: The paper adapts frameworks developed for large organisations and national systems into a graduated standard for small providers, and applies it reflexively.

Keywords: evidence-based education; training evaluation; impact evaluation; education technology; randomised controlled trials; Kirkpatrick; Kenya

1. Introduction

1.1 An evidence problem in education technology and training

Education technology is sold on promises of improved learning. The question that matters to a learner, a ministry or a funder is whether those promises have been tested, by whom, and against what. The 2023 Global Education Monitoring Report put the problem plainly. It noted that much of the evidence on education technology is produced by vendors, that only 7% of education technology companies in the United Kingdom had conducted randomised controlled trials, and that in a survey of 17 US states only 11% of teachers and administrators had asked for peer-reviewed evidence before adopting a product. It also recorded a case in which studies funded by a major publisher contested independent analysis that had found its products had no impact (UNESCO, 2023). The report's summary judgement was that not all change constitutes progress, and that decisions about technology should prioritise learner needs after assessing whether it is appropriate, equitable, evidence-based and sustainable.

The weakness is not confined to school technology. In workforce and professional training, programmes are commonly judged by whether participants enjoyed them. That is easy to measure and, as Section 3 shows, only weakly related to whether anything changed. For a small provider in a low-resource setting the difficulty is compounded. Rigorous evaluation

appears to require resources the provider does not have, which tempts it either to say nothing or to claim more than it can support. Funders face the mirror-image difficulty. They want evidence of impact but cannot easily distinguish a provider that has measured its work honestly from one that has produced a persuasive report.

1.2 The gap in the literature

Two bodies of work address this problem and they rarely meet. The training-evaluation literature, descending from Kirkpatrick's four levels, tells practitioners what to measure but is relatively silent about how strong the evidence from a given design is (Kirkpatrick & Kirkpatrick, 2006). The evidence-hierarchy literature, from experimental economics and education policy, tells decision makers how to grade evidence but is designed for large programmes and national systems that can afford randomisation (Glennerster & Takavarasha, 2013; Gertler et al., 2016). Between them lies the practical situation of most providers, which are small, deliver programmes of modest size, and must decide what is worth measuring and what claims their data support. Few frameworks address that situation directly.

1.3 Contribution and structure

This paper proposes a proportionate evidence standard for small providers. It makes three contributions. First, it integrates the four levels of training evaluation with the tiers of evidence used in United States education law into a four-rung evidence ladder that ties each level of measurement to the claims it can support. Second, it tests the logic that the ladder implies, that programmes should teach to the learner's level, support teachers and be measured, against experimental evidence from Kenya, Sub-Saharan Africa and elsewhere. Third, it specifies a minimum reporting standard, six testable propositions, and a reflexive assessment of the provider with which the first author is associated.

Section 2 describes the method. Section 3 develops the theoretical foundations and the framework. Section 4 reviews what experimental evidence says works. Section 5 considers how a small provider can measure what changes. Section 6 treats threats to validity and integrity. Section 7 sets out the reporting standard. Section 8 states propositions. Section 9 applies the framework to Instracta. Sections 10 to 12 discuss implications, limitations and conclusions.

2. Method

2.1 Type of review and its limits

This is a theory-building narrative review. It is not a systematic review conducted to a PRISMA protocol and reports no screening counts. Its aim is to synthesise evidence from two literatures

into a usable standard and to derive propositions. The principal weakness of the approach is selection: without a protocol, a reviewer may choose evidence that supports a preferred view. The steps below are intended to limit that risk. They also allow a reader to reproduce the search and to test whether the conclusions depend on the sources chosen.

2.2 Search, inclusion and appraisal

Searches were conducted in October 2026 using a general web search engine and repository records of publishers and research networks (including the American Economic Association, the National Bureau of Economic Research, the World Bank Open Knowledge Repository, EdTech Hub, Open Development and Education, and IDEAS/RePEc). Strings combined evaluation terms with regional and design terms, for example Kenya randomised evaluation reading, teacher professional development meta-analysis Sub-Saharan Africa, tiers of evidence ESSA, training transfer meta-analysis, and publication bias. Appendix A lists the principal strings.

Sources were included if they reported experimental or quasi-experimental estimates of learning or training effects in low- or middle-income countries, reviewed such evidence systematically, or made a substantive theoretical contribution to training evaluation, evidence grading or the interpretation of effect sizes. Vendor reports and promotional material were excluded as evidence. Appraisal used the hierarchy that the paper itself proposes, with meta-analyses and systematic reviews ranked above single randomised studies, above quasi-experimental and descriptive studies, above theory. Effect sizes are reported as the authors state them, in standard deviations, and are not pooled, because outcomes, populations and time horizons differ. For several sources only the abstract or catalogue record could be accessed, and statements about them are limited to what those records report. Statements derived from secondary reports are marked.

2.3 Positionality

The first author is a practitioner who founded a Kenyan learning provider and who leads a learning management service at a university. Practitioner experience informed the choice of questions and the pragmatism of the standard proposed. It also creates an interest in the conclusions, declared in the closing statements. The paper states its claims in a form that can be checked against sources and presents counter-evidence where it exists.

3. Theoretical foundations and the evidence ladder

3.1 What counts as evidence of impact

Impact is a causal claim: that a programme changed an outcome relative to what would have happened without it. The central problem of impact evaluation is therefore the construction of a credible counterfactual. Randomised designs achieve this by assigning participants by chance, so that on average the groups differ only in the programme. Quasi-experimental designs approximate it using matching, discontinuities or statistical control, at the cost of assumptions that cannot be fully tested (Shadish et al., 2002; Gertler et al., 2016). Before-and-after comparisons without a comparison group cannot separate the programme from maturation, from other events or from the tendency of those who select a programme to improve anyway.

Two distinctions matter. Internal validity concerns whether the study identifies the effect in its own setting. External validity concerns whether the result applies elsewhere. A well-run randomised trial has strong internal validity and, depending on the setting, may have weak external validity. Pritchett and Sandefur (2015) examined the trade-off using microcredit, migration and education policies and found that, by mean-squared error, non-experimental evidence from the same context performed better than experimental evidence from a different one. Vivalt (2020), in a study of generalisability across impact evaluations, documented substantial heterogeneity in results across settings. Both argue that context-specific evidence has a value that rankings of study designs can obscure, an argument that favours small providers evaluating their own programmes even with modest designs.

3.2 Kirkpatrick's four levels and their critics

The dominant framework in training evaluation is Kirkpatrick's four levels: reaction, learning, behaviour and results (Kirkpatrick & Kirkpatrick, 2006). Reaction asks whether participants found the training engaging and relevant. Learning asks whether knowledge, skill or attitude changed. Behaviour asks whether participants apply what they learned in the workplace or classroom. Results asks whether the organisational or community outcomes the training was meant to move did move. Phillips (1997) added a fifth level, return on investment.

The framework has been criticised on three grounds. Alliger and Janak (1989), in a review thirty years after the model was introduced, questioned the assumption of causal linkages and positive intercorrelations between the levels. Holton (1996) argued that the model is a taxonomy of outcomes and not a fully specified model of how training produces them, because it omits the individual and environmental influences that determine whether learning translates into performance. Bates (2004) added that its neglect of contextual factors can lead to the misattribution of failure to the training itself. Empirical work supports the scepticism. A meta-analysis of the relations among training criteria found that they were related, but modestly (Alliger et al., 1997), and a meta-analysis of trainee reactions found that reactions predicted pre-to-post change in motivation ($\beta = .51$) and self-efficacy ($\beta = .24$), were more sensitive than learning outcomes to trainees' perceptions of course characteristics, and were more strongly related to outcomes in courses with a high level of technology (Sitzmann et al., 2008). Reaction

data are therefore informative about the experience of a course, especially in technology-mediated settings, but are not a substitute for measures of learning and application.

At the same time, there is positive evidence that organisational training can work. A meta-analysis by Arthur, Bennett, Edens and Bell (2003) found that training was on average effective across reaction, learning, behaviour and results criteria, with effect sizes varying with the training method and with the criterion used. Taken together, the literature suggests that the four levels are a useful list of what to measure but not a causal chain that can be assumed, and that evaluations should measure at least learning and, where feasible, behaviour, and should not infer the higher levels from the lower.

3.3 Tiers of evidence

On how much confidence evidence deserves, we draw on the tiers defined in the Every Student Succeeds Act of the United States and in the guidance that implements it. Tier 1, strong evidence, requires a well-designed experimental study meeting What Works Clearinghouse design standards without reservations. Tier 2, moderate evidence, requires a well-designed quasi-experimental study, or an experimental study with high attrition that meets standards with reservations. Tier 3, promising evidence, requires a well-designed correlational study with statistical controls for selection bias, or a quasi-experimental or experimental study that does not demonstrate baseline equivalence. Tier 4, demonstrates a rationale, requires only an intervention based on a well-specified logic model that is itself based on rigorous research, with a planned or ongoing evaluation (Center for the Analysis of Data in Education Research [CADRE], n.d.).

The tiers were designed for United States schools and are not a Kenyan or African standard. Their value here is that they are explicit and graduated, do not demand randomisation from a provider who cannot yet afford it, and refuse to treat a logic model as proof. A provider can state which tier its evidence meets, and a funder can check.

3.4 The evidence ladder

We combine the two frameworks in an evidence ladder with four rungs. Each rung specifies what the provider does, which levels of Kirkpatrick's model it can credibly measure, and which claim its data can support. Figure 1 depicts the ladder.

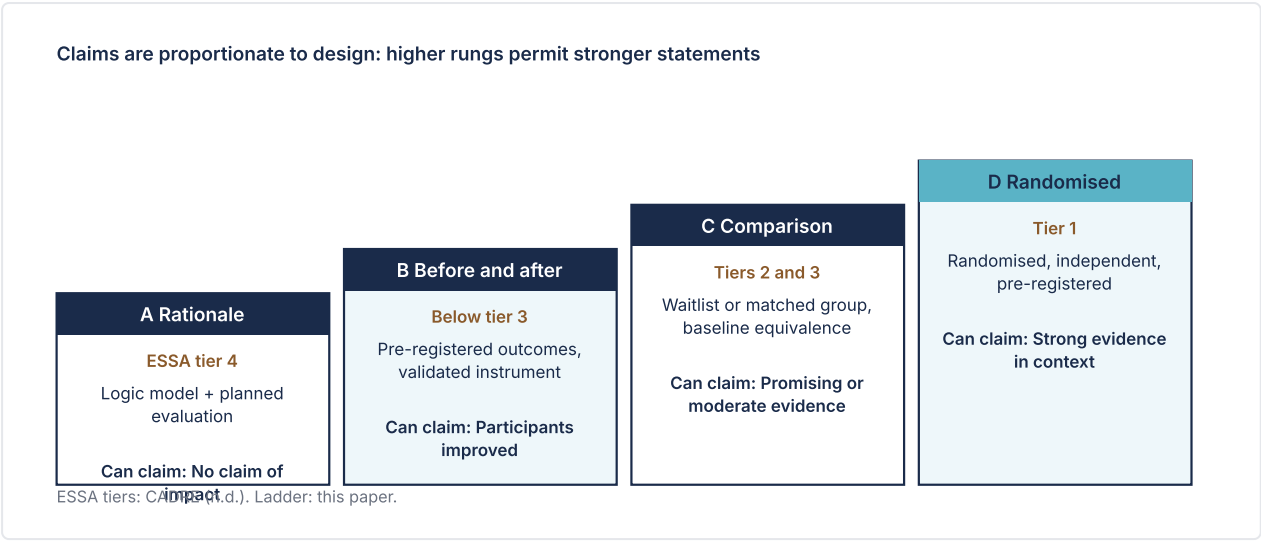


Figure 1. Evidence ladder for small providers. Each rung corresponds to a design a small provider can run, an ESSA evidence tier, and the claim that the evidence supports. Providers should state their rung publicly and commit to the next with a date.

Table 1. Evidence ladder: provider actions, measured levels and permitted claims

Rung	ESSA tier	What the provider does	Kirkpatrick levels measurable	Permitted claim
A. Rationale	Tier 4	Publishes a logic model linked to named research; collects reaction and learning data; plans an evaluation	1, and 2 descriptively	The design is consistent with research. No claim of impact.
B. Before and after	Below tier 3	Pre-registers outcomes; measures learning before and after with a validated instrument; reports attrition; disaggregates	1 and 2; 3 at follow-up	Participants improved on the measured outcome. Cause not established.
C. Comparison	Tiers 2 and 3	Adds a comparison group (waitlist or matched), checks baseline equivalence, uses statistical controls; independent analysis	1 to 3; 4 where data exist	Promising or moderate evidence of effect, with stated limits.
D. Randomised	Tier 1	Randomises participants or sites (waitlist lottery is often feasible); pre-registers; partners with a research body	1 to 4	Strong evidence of effect in this context.

The ladder has three virtues. It makes claims proportionate to design: a provider on rung B says that participants improved, not that the programme caused the improvement. It gives a small provider a legitimate starting point, since rung A requires thought and honesty and not money. And it is cumulative, because the instruments and data collected at rung B are the baseline for rung C. Waitlist randomisation deserves emphasis. When more people apply than a funded programme can serve, assigning places by lottery is a fair allocation rule as well as a research design. The lottery in Muralidharan et al. (2019) worked in exactly this way, with winners receiving a voucher and losers forming the comparison group.

3.5 Logic models as the link between design and measurement

A logic model states how inputs and activities are expected to produce outputs, outcomes and impact, and why. Its value is that it makes assumptions testable. A programme that expects coaching to change teachers' practice, and practice to raise learning, can measure each link. A programme that cannot state the links cannot tell which failed. Figure 3 in Section 5 places the four levels along the results chain.

3.6 Relation to the companion papers

This paper is the second of three working papers addressing the principles on which Instracta's work is organised. The first paper argues that access should be assessed at four gates, the last of which, learning, depends on the evidence standard developed here. The third paper addresses local provision: language, pedagogy and institutions. The three are intended to be read together. Evidence without access describes programmes that reach few, and evidence without local fit describes effects that may not travel. The evidence ladder is the mechanism by which the claims of the other two papers can be tested.

4. What the experimental evidence says works

A proportionate standard should be anchored in what the best studies have shown works, so that a provider's logic model does not rest on fashion. Six findings recur across the experimental literature. The first four concern instruction and teachers, the fifth technology, and the sixth how to read effect sizes.

4.1 Matching instruction to the learner's level

Banerjee, Cole, Duflo and Linden (2007) reported two randomised experiments in India. In the first, young women from the local community taught basic literacy and numeracy to children lagging behind in government schools in urban slums, and average test scores in treatment schools rose 0.14 standard deviations in the first year and 0.28 in the second. In the second, fourth graders received two hours a week of shared computer time playing mathematics games, and their scores rose 0.35 standard deviations in the first year and 0.47 in the second. Gains persisted at least a year after the programmes ended, and classmates who experienced smaller classes alone showed no gains, which led the authors to conclude that adding resources may not be enough to improve outcomes. Muralidharan, Singh and Ganimian (2019) found that a personalised, technology-aided after-school programme raised mathematics scores by 0.36 and Hindi scores by 0.22 standard deviations over about 4.5 months for students who won a lottery for a place, with gains that were proportionally larger for academically weaker students because control-group students with weak preparation learned at a rate close to zero.

In Kenya, Duflo, Dupas and Kremer (2011) compared 61 schools that randomly assigned students to first-grade classes with 60 schools that tracked students by initial achievement.

Students in tracking schools scored 0.14 standard deviations higher after 18 months, the advantage persisted a year after the programme ended, and the authors interpreted the effect as arising from teachers being able to pitch instruction at a level suited to their students. Evans and Popova (2016) found that, among six systematic reviews with divergent conclusions, one of two types of programme recommended consistently was teaching tailored to students' learning levels, whether delivered by teachers or adaptive software. The finding is robust across settings and delivery modes, and it is the strongest single argument for diagnostic assessment and adaptive pathways in digital learning.

4.2 Sustained, in-classroom support for teachers

Piper, Simmons Zuilkowski and Mugenda (2014) reported on a randomised trial of the PRIMR Initiative in more than 400 Kenyan schools in three counties. The package aligned curriculum and teacher practice with current research, provided ongoing instructional support and classroom observation, and supplied basic materials and books in English and Kiswahili. It improved oral reading fluency in both languages, with gains in grade 1 in formal and non-formal schools and in grade 2 in non-formal schools. The authors concluded that in-classroom teacher support was important. In a further analysis of the PRIMR randomised trial, Piper, Zuilkowski, Dubeck, Jepkemei and King (2018) compared combinations of teacher professional development and coaching, student textbooks and structured teachers' guides, and found that two of three combinations had statistically significant positive effects, that the most effective combined professional development, coaching, one-to-one student books and structured lesson plans, and that the most expensive option was also the most cost-effective because its added impact outweighed its cost.

Conn (2017), in a meta-analysis of 56 articles covering 66 experiments and quasi-experiments in Sub-Saharan Africa, found that programmes changing teacher pedagogy or classroom instruction had effect sizes about 0.30 standard deviations larger than all other programme types combined, with limited evidence that adaptive instruction and teacher coaching were especially effective. Evans and Popova (2016) found that the second consistently recommended type of programme was individualised, repeated teacher training linked to a specific task or tool. The common thread is that support is sustained and tied to practice, not delivered once as general training.

4.3 Teacher knowledge as a binding constraint

Bold and colleagues (2017) used nationally representative surveys from seven Sub-Saharan African countries, covering close to 40% of the region's population, and found that students received about two hours and fifty minutes of teaching a day, just over half the scheduled time, that many teachers did not fully master the curriculum they taught, that basic pedagogical knowledge was limited, and that effective teaching practices were uncommon. Using variation within students and within teachers, they found large positive effects of teachers' content and pedagogical knowledge on student achievement. For a training provider the finding suggests

that programmes aimed at teachers and trainers have a plausible route to impact, and that their evaluation should measure changes in knowledge and in classroom practice as well as in learner outcomes.

4.4 Complementarities: inputs, incentives and instruction

Mbiti and colleagues (2019) randomised interventions in 350 Tanzanian schools and found that unconditional school grants had no measurable effect on test scores after two years, that teacher incentives tied to student performance had some positive effect, and that the two combined had significant effects larger than the sum of the separate effects. They interpret the result as complementarity between inputs and incentives. The lesson for evaluation is that the effect of a component can depend on what surrounds it, so that a programme evaluated as a bundle may not tell a provider which part matters, and the same component in a different bundle may behave differently. Piper et al. (2018) make the same point for the components of a literacy package. Evaluations that vary components, even in simple ways, are more informative than those that test one bundle.

4.5 Technology: a carrier of pedagogy, evaluated like any other

Escueta, Nickow, Oreopoulos and Quan (2020) reviewed randomised and regression-discontinuity studies of technology in education across four categories: access to technology, computer-assisted learning, technology-enabled behavioural interventions and online learning. They concluded that technology can help, and in some cases hinder, learning. Low-technology designs can work. Angrist, Bergman and Matsheng (2022) found in a randomised trial in Botswana that SMS messages and phone calls to parents during school closures improved learning by 0.12 standard deviations at low cost, and Aker, Ksoll and Lybbert (2012) found that an adult education programme in Niger in which students learned to use basic mobile phones raised scores by about 0.19 to 0.26 standard deviations compared with conventional classes, with the mathematics advantage persisting seven months after classes ended. Hennessy and colleagues (2022) presented what they described as the first systematic review of technology-mediated teacher professional development in low- and middle-income countries, noting that professional development in these contexts is often limited and short-lived and that technology may strengthen both formal programmes and informal peer learning.

The 2023 Global Education Monitoring Report adds two cautions. Systematic reviews over two decades find small to medium positive effects compared with traditional instruction, but evaluations often cannot isolate the effect of technology from additional instruction time or teacher support, and in Peru a distribution of more than one million laptops had no positive effect on learning (UNESCO, 2023). The literature supports a consistent position: technology is a means of delivering instruction, practice, feedback or communication, and it should be evaluated for what it delivers, not as a category. Figure 2 summarises the effect sizes reported in the randomised studies cited in this section.



Figure 2. Reported learning effects, in standard deviations, from selected randomised evaluations. Outcomes, populations and time horizons differ, so the figure shows orders of magnitude and not a ranking. Sources: Banerjee et al. (2007); Muralidharan et al. (2019); Duflo et al. (2011); Aker et al. (2012); Angrist et al. (2022).

4.6 Reading effect sizes

An effect size is only meaningful relative to a benchmark. Kraft (2020) argued that the conventional thresholds of 0.2, 0.5 and 0.8 standard deviations, developed for laboratory psychology, are inappropriate for education interventions evaluated on standardised achievement in causal studies, and proposed that effects below 0.05 be regarded as small, those from 0.05 to below 0.20 as medium, and those of 0.20 or greater as large. Hill and colleagues (2008) similarly proposed benchmarking effect sizes against typical annual achievement gains and against policy-relevant gaps. The upshot is that a provider who reports a gain of 0.1 standard deviations is reporting a medium effect by Kraft's standard, and that the same number can be small or large depending on the measure, the population and the time over which it accrued. Effect sizes should be reported with confidence intervals, the outcome measure and the duration.

4.7 Context, scale and heterogeneity

Results travel imperfectly. Vivalt (2020) documented substantial heterogeneity across the settings in which impact evaluations were conducted, and Pritchett and Sandefur (2015) showed that local non-experimental evidence can outperform experimental evidence from elsewhere. Banerjee and colleagues (2017) reviewed the challenges of moving from a successful proof of concept to a scalable policy, including the way effects can fall when programmes are delivered by governments and at larger scale. The most defensible reading of the literature for a Kenyan provider is therefore that the findings in Sections 4.1 to 4.5 are good

priors for what is likely to work, particularly targeted instruction and sustained teacher support, but that effect sizes should not be borrowed. They should be measured locally.

4.8 What the evidence does not tell us: adult and workforce learning

The experimental evidence reviewed above comes overwhelmingly from school-age learners and basic skills. The Kenyan trials concern early-grade reading and numeracy, the Indian trials concern middle-school mathematics and language, and the Niger study concerns adult basic literacy and numeracy. Evidence on whether online or blended training improves workplace skills, employability or income among young adults in Africa is much thinner. The organisational training meta-analysis of Arthur et al. (2003) and the transfer literature (Blume et al., 2010) are drawn mainly from high-income settings and workplaces. This matters because the programmes of many digital course providers, including the provider discussed in Section 9, are aimed at young adults and professionals. A provider in this market cannot cite the school-based evidence as proof that its programmes work. It can cite it as support for design principles, and it should treat the lack of direct evidence as an opportunity: a well-designed evaluation of a workforce programme in Kenya would be a contribution to the field as well as to the provider's own knowledge.

5. Measuring what changes in practice

5.1 Measurement points along the results chain

A provider cannot measure everything. The question is where measurement is most informative given a programme's logic. Figure 3 shows a logic model from inputs to impact and the four levels of Kirkpatrick's framework as measurement points along the chain. The figure includes a learning loop, because the purpose of measurement is to change the design as well as to report it.

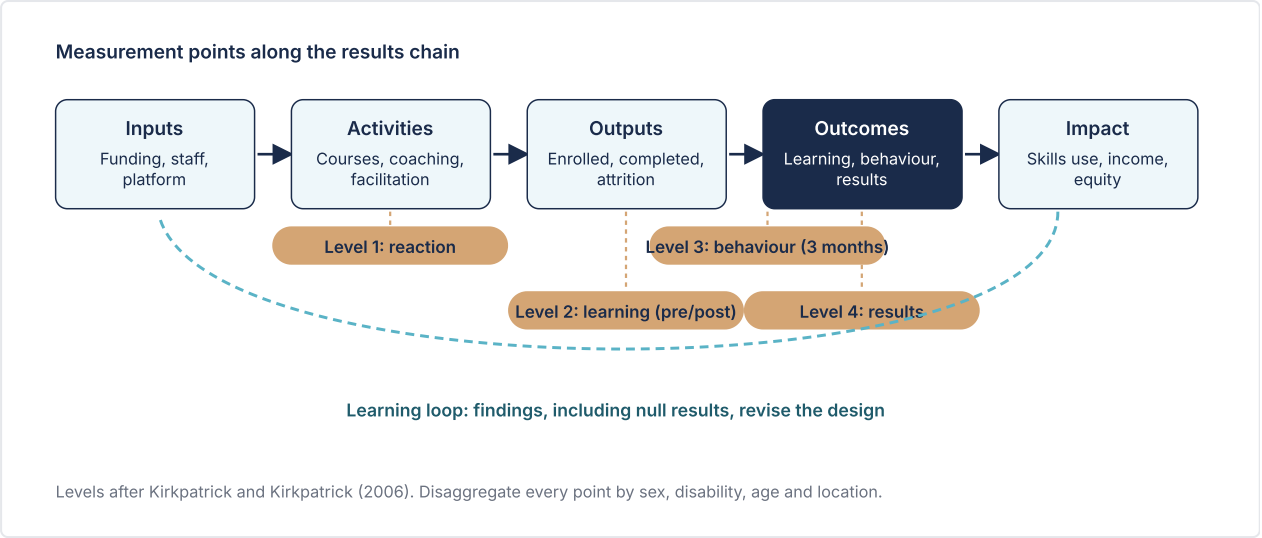


Figure 3. Logic model with Kirkpatrick measurement points and learning loop. Every measurement point should be disaggregated by sex, disability, age and location. Null and negative findings feed back into the design in the same way as positive ones.

5.2 Instruments by level

Table 2 lists instruments suited to a small provider at each level. The aim is to use the least costly instrument that is valid for the purpose. At the learning level a short, validated, pre-specified assessment administered before and after is more informative than satisfaction ratings. At the behaviour level, structured observation, work samples, supervisor reports or learner diaries at a fixed interval after training are the usual choices. At the results level, the choice depends on the programme: learner outcomes in a school, job performance in an organisation, or income and employment in a skills programme.

Table 2. Measurement instruments by level for small providers

Level	Question	Example instruments	Typical timing
1 Reaction	Was the experience engaging, relevant and usable?	Short post-course survey; usage and drop-out data; accessibility feedback	End of course
2 Learning	Did knowledge, skill or confidence change?	Pre-specified pre and post assessments; practical task with rubric	Before and after; follow-up at 1 to 3 months
3 Behaviour	Do participants apply what they learned?	Observation checklist; work samples; supervisor or peer reports	1 to 6 months after
4 Results	Did the outcomes the programme targets move?	Learner outcomes; organisational indicators; employment and income records	3 to 12 months after

5.3 Why behaviour and results are the hard levels: transfer of training

The levels most valued by funders are the hardest to measure and to influence. The literature on transfer of training explains why. Baldwin and Ford (1988) reviewed the field and organised the determinants of transfer into trainee characteristics, training design and the work environment. Blume and colleagues (2010) updated the evidence in a meta-analytic review of the predictors of transfer. The consistent message is that the work environment, including supervisor support and the opportunity to apply what was learned, shapes whether training changes practice, and that training design alone cannot guarantee transfer. A provider that evaluates only inside the course will therefore miss the factors that most affect results, and a provider that measures results without attending to the work environment may wrongly conclude that a sound programme failed. Practical consequences follow. Evaluations should record whether participants had the opportunity and support to apply what they learned, and programmes should design for transfer by involving supervisors and by including application tasks tied to the learner's real work.

5.4 Comparison designs a small provider can run

Three comparison designs are within reach of most providers. The first is the waitlist lottery described in Section 3.4, which assigns the places in an oversubscribed programme by chance

and uses those who wait as the comparison group. The second is matched comparison, in which participants are compared with non-participants selected to resemble them on baseline measures; it is weaker, because unmeasured differences remain, but the ESSA tiers recognise it where baseline equivalence is demonstrated. The third is staggered roll-out, in which cohorts start at different times and the later cohorts serve as comparisons for the earlier ones while they wait. Each can be run alongside normal delivery. Each requires that outcomes be pre-specified and measured in the same way for both groups, and that attrition be tracked. Randomised designs should obtain ethical review and informed consent, and fair allocation rules should be explained to participants in advance.

5.5 Cost-effectiveness and the cost of evidence

Impact without cost is incomplete evidence. Piper et al. (2018) showed that the most expensive combination of components can be the most cost-effective, and McEwan (2015), in a meta-analysis of 77 randomised experiments with 111 treatment arms, provides a wide base for comparing intervention types in primary schools. For a provider, the practical measure is the cost per participant who completes and, where outcomes allow, the cost per unit of measured learning gain. Reporting both allows funders to compare programmes on a common basis and discourages the optimisation of the cheaper but less informative denominator, enrolments. The cost of evaluation also deserves attention. A small provider can expect that a rung B evaluation will cost a modest share of programme cost, mainly in instrument development and analysis time, and that a rung D evaluation will cost materially more. These magnitudes have not been established for Kenya, and a useful research task is to estimate them.

5.6 Learning analytics and data ethics

Digital programmes generate data on activity that can supplement evaluation. Siemens and Long (2011) described the promise of analytics for learning and education, and Slade and Prinsloo (2013) set out ethical issues and dilemmas in its use, including consent, privacy, the potential for labelling learners and the responsibility to act on what the data reveal. Log data are a cheap source of reaction and behaviour indicators, but they measure activity, not learning, and can mislead: a learner who is inactive may be learning offline, and a learner who is active may not be learning. In Kenya, the Data Protection Act of 2019 gives individuals rights over their personal data and places obligations on data controllers and processors (Republic of Kenya, 2019). Evaluation data are personal data. A provider should collect only what the evaluation needs, obtain informed consent for evaluation use, store and share data securely, and de-identify data before analysis by third parties.

5.7 Statistical power: what small cohorts can and cannot detect

Small providers face a constraint that is rarely stated: cohorts are small, and small cohorts can detect only large effects. The minimum detectable effect in standard deviations for a two-arm

comparison with equal groups, a two-sided test at the 5% level and 80% power is approximately 2.8 multiplied by the square root of two divided by the number of participants per arm. With 60 participants per arm the minimum detectable effect is about 0.51 standard deviations. If a baseline measure that explains half the variance in the outcome is included as a covariate, the minimum detectable effect falls to about 0.36. These figures are illustrative arithmetic and not a substitute for a power calculation tailored to a design (Glennerster & Takavarasha, 2013).

The comparison with Figure 2 is sobering. Most of the effects reported in the randomised studies cited in this paper lie between 0.12 and 0.47 standard deviations, and the median is well below the minimum detectable effect of a single cohort of 120. A provider that runs a single small cohort and finds no statistically significant difference has not shown that the programme does not work. A provider that finds a large and significant effect in a single small cohort should suspect that the estimate is inflated. The practical remedies are to include a baseline measure, to pool cohorts over time using the same instruments and design, to report confidence intervals so that readers can see the range of effects consistent with the data, and to avoid claims of no effect from underpowered tests. Pooling also argues for stable instruments and procedures from the first cohort, which is a further reason to pre-specify them.

5.8 Mixed methods and implementation evidence

Numbers show whether outcomes changed. They rarely show why. Process evaluation, which documents what was actually delivered, to whom, with what fidelity and what adaptations, is necessary to interpret outcomes: a null result may reflect a poor design or a poorly delivered one, and a positive result may depend on features that cannot be replicated. Qualitative evidence from learners, facilitators and employers, collected systematically and reported with its limits, can identify mechanisms and unintended effects. Where randomisation is impossible, regression and structural equation modelling can estimate associations between programme features, learner characteristics and outcomes, but they estimate association and not causation, and should be reported in line with rung B or C of the ladder. A well-designed evaluation at any rung combines at least one outcome measure with documented implementation, because the pair makes the result usable by someone else.

6. Threats to validity and to integrity

6.1 Threats to internal validity

Small providers are exposed to a recognisable set of threats. Selection bias arises when the people who choose a programme differ from those who do not, and is the main reason that before-and-after results overstate effects. Attrition biases results if those who drop out differ

from those who stay, which is common in online courses with high drop-out; the comparison of participants who complete with a comparison group that does not is thus a comparison of different populations. Novelty and attention effects can inflate early results. Maturation and history can produce change that the programme did not cause. Measurement problems arise when the programme teaches the content of the test, when instruments are not validated for the population or language, or when participants know their answers will be used to judge the programme. Each can be reduced by design, and each should be reported.

6.2 Threats to integrity: publication bias, vendor evidence and funder pressure

A second set of threats concerns what is reported. Franco, Malhotra and Simonovits (2014) tracked a cohort of social science experiments from registration to publication and found that null results were far less likely to be published, or even written up, than strong results, so that the published literature overstates effects. Nosek and colleagues (2015) argued for an open research culture, with pre-registration, transparency and data sharing, to reduce the distortion. In education technology, the Global Education Monitoring Report notes that much evidence comes from vendors (UNESCO, 2023). A provider evaluating its own product has an incentive to report favourable results, and a funder who wants a success story adds to it. The remedy is structural and not moral: pre-registration of outcomes, an independent party who holds the data and signs the analysis, and a commitment to publish all results.

Independence can be achieved at modest cost. A university research unit, a development-economics group or a qualified independent evaluator can review the design, hold the data and sign the analysis. The role is distinct from that of an advisory board, and its occupant should not be responsible for delivery or paid on results. The 2023 Global Education Monitoring Report recommends that governments establish independent bodies to evaluate education technology (UNESCO, 2023), and the same principle applies to individual providers.

6.3 Ethics of evaluation in low-resource settings

Evaluation raises ethical questions that are sharper where participants are poor and providers are powerful. Participants must give informed consent that is genuinely voluntary: a learner who believes that declining to take part in an evaluation will cost her a place in the programme is not free to refuse. Allocation by lottery is defensible when places are scarce, but participants should be told how places are allocated, and those on the waitlist should be offered the programme afterwards where possible. Evaluation should not impose costs on participants, such as time or data for surveys, without compensation. Data should be collected only for stated purposes, and participants should know who will see them. The Data Protection Act of 2019 provides a legal floor (Republic of Kenya, 2019), and an ethics review by a research ethics committee is a sensible practice for any evaluation that involves randomisation or sensitive data. Finally, evaluation findings should be shared with the participants and the communities

from which they came. Evidence that benefits only funders and academics is a form of extraction.

7. A minimum reporting standard

Whatever the rung, a provider should meet a minimum reporting standard. The twelve items below are drawn from common practice in impact evaluation, from the evidence guidance cited above and from the threats in Section 6. They are designed to be checkable by a funder from a short evaluation report.

1. State the outcome to be measured, and how, before data collection, and register it publicly.
2. Name the instrument and say whether it was validated for the population and language.
3. State the rung of the evidence ladder and the corresponding ESSA tier, and say what claim the design supports and what it does not.
4. Report the number enrolled, the number who completed, and the number measured at endline, so that attrition is visible.
5. Report effect sizes, not only whether a result was statistically significant, with confidence intervals, and interpret them against a stated benchmark.
6. Disaggregate results by sex, disability status, age and location.
7. Describe the comparison, or state that there was none, and show baseline equivalence where a comparison exists.
8. Report the cost per participant and, where outcomes allow, the cost per unit of learning gain.
9. Identify who conducted the analysis and whether they were independent of the delivery team.
10. Report whether participants had the opportunity and support to apply what they learned, where behaviour or results are claimed.
11. Publish null and negative findings with the same prominence as positive ones.
12. Protect participant data in line with the Data Protection Act, 2019 of Kenya, and obtain informed consent for evaluation use.

The standard is deliberately minimal. A provider that meets it has said what it measured, how and with what limits, and has not claimed more than the design supports. A funder that requires it can compare providers on a common basis and can reward honesty about null results, which are the strongest signal that an evaluation is not being managed for effect.

8. Propositions

The framework and evidence imply propositions that can be tested. They are stated as directional claims and are offered as a research agenda.

Table 3. Propositions derived from the evidence ladder framework

No.	Proposition	Basis in the evidence reviewed
P1	Providers that report only reaction data will overstate impact relative to providers that measure learning with a validated instrument.	Sitzmann et al. (2008); Alliger & Janak (1989); Alliger et al. (1997)
P2	Before-and-after estimates of learning gain for self-selected participants will exceed estimates from waitlist or matched comparisons of the same programme.	Shadish et al. (2002); Muralidharan et al. (2019); Pritchett & Sandefur (2015)
P3	Programmes that combine content with diagnostic assessment and instruction matched to learners' levels will show larger gains than programmes that deliver the same content uniformly.	Banerjee et al. (2007); Muralidharan et al. (2019); Duflo et al. (2011); Evans & Popova (2016)
P4	Training programmes that include supervisor involvement and application tasks will show larger behaviour change at follow-up than programmes that do not.	Baldwin & Ford (1988); Blume et al. (2010); Piper et al. (2014)
P5	Gains measured on the same programme will be larger in the first pilot than at scale.	Banerjee et al. (2017); Vivalt (2020)
P6	Providers with an independent evaluator who holds the data and commits to publish will report a smaller share of positive findings than providers without.	Franco et al. (2014); Nosek et al. (2015); UNESCO (2023)

P1 to P6 are untested for Kenyan training providers to the authors' knowledge. P6 would require a sample of providers and cooperation from funders.

9. Reflexive application: Instracta eLearning Solutions

9.1 Purpose and caution

This section applies the framework to the provider with which the first author is associated. The exercise is reflexive, and its limits are those of any self-assessment by an interested party. It is therefore confined to stating the provider's rung, its commitments and the evidence it has not yet produced. It does not claim that the provider's programmes improve learning.

9.2 Position on the ladder

On the evidence the provider has documented to date, its programmes sit at rung A, with some descriptive data approaching rung B. Where it reports results from past work, those results should be read as descriptive and not as causal. This is not unusual for a small provider and it is a reasonable starting point, provided that the next rung is committed to publicly and dated. Table 4 states the provider's commitments against the ladder and the reporting standard.

Table 4. Provider commitments against the ladder and reporting standard

Commitment	Rung or item	Status in October 2026
Publish a logic model for each initiative that names the research on which it rests	Rung A	To be published
Move the first sponsored cohort to rung B with pre-specified outcomes and a validated instrument	Rung B; items 1, 2, 4	Planned; dependent on funding
Move at least one programme to rung C using a waitlist comparison	Rung C; items 7, 11	Planned; dependent on funding and cohort size
Appoint an independent evaluator who does not deliver the programme	Item 9	An evaluator has been proposed; appointment terms to be confirmed
Publish cost per participant and cost per completer	Item 8	To be published with the first cohort
Publish findings whether or not they flatter the organisation	Item 11	Commitment stated

9.3 What the exercise shows

As with the first paper, the exercise produced a short, checkable list of commitments and gaps. It also illustrates an asymmetry. The commitments that cost least, the logic model and the publication policy, can be met immediately. The commitments that matter most, a comparison design and an independent analysis, depend on funding and on cohort size. That asymmetry is itself an argument for funders to underwrite evaluation, because a provider cannot produce the evidence funders ask for without the means to do so. It is also an argument for sequencing: a provider with limited resources should complete the inexpensive commitments first and publish them, so that the funder can see both the commitments met and the evidence still to come.

10. Discussion and implications

10.1 For providers

For providers, the most useful result is the licence to be modest. A rung A provider that says exactly what it knows is in a stronger position, with funders who read carefully, than one that claims impact on the strength of satisfaction data. The ladder converts evidence from a binary, proven or not, into a path. A provider can state its rung, commit to the next, and report progress. It should resist the temptation to borrow effect sizes from the studies in Section 4. Those studies justify the design, not the claim.

10.2 For funders

For funders, the framework suggests due-diligence questions. At which rung is the evidence, and what claim does it support? Were outcomes pre-registered? Who analysed the data, and are they independent? What are the attrition and the disaggregated results? What is the cost per completer? Were null results reported? It also suggests a funding approach. Funders who require evidence should fund it: a modest evaluation line in grants, a shared pool of validated instruments, and a framework that rewards honest reporting of null results. Funders can also reduce the burden on small providers by accepting a common reporting standard instead of requiring each to produce a bespoke evaluation.

10.3 For policy

For policy makers, the 2023 Global Education Monitoring Report recommends establishing independent bodies to evaluate education technology, piloting in ways that reflect the total cost of ownership, and transparency in spending and contracts (UNESCO, 2023). The evidence ladder is compatible with that recommendation and could be adapted by a Kenyan education or skills agency as a procurement filter, so that tools and training programmes bought with public money state the rung and tier of their evidence. A national evidence standard for education technology would also reduce the burden on schools and colleges, which lack the capacity to assess vendor claims.

A related policy question concerns the use of public data. Kenya's administrative data on training enrolment, certification and employment, where they exist and can be linked with consent and with protection for privacy, could allow providers and researchers to estimate longer-term results at lower cost than bespoke surveys. Such linkage would also strengthen the results level of Kirkpatrick's model, which is the level that small providers find hardest to measure on their own. The conditions are that the data are lawfully accessed, that privacy is protected, and that findings are shared with those who contributed the data.

10.4 For research

Three questions would most advance the field. First, what does a rung B or rung C evaluation cost in Kenya, by programme type, and what share of programme budgets is that? Second, how do estimates differ between the before-and-after, matched and randomised designs when applied to the same programme (P2)? Third, which instruments are valid and reliable for

Kenyan adult and workforce learners across languages? A shared item bank of validated, openly licensed instruments would lower the cost of rungs B and C for all providers.

10.5 A worked illustration of the ladder

To make the ladder concrete, consider a hypothetical sponsored cohort of an employability course. The numbers are invented for illustration. Suppose that 120 eligible young people apply for 60 sponsored places. The provider publishes a logic model stating that the course is expected to raise a specified set of workplace-communication skills, and registers the outcome: the score on a 40-item assessment, with a pre-specified rubric for a practical task. It allocates the 60 places by public lottery and invites the 60 unselected applicants to complete the same assessments at the same times, promising them places in the next cohort. Both groups complete the assessment at baseline and eight weeks later, and both are asked about their application of the skills three months later. An independent evaluator holds the data, analyses it with the baseline score as a covariate and signs the report.

What can the provider claim? With 60 per arm and a baseline covariate, the design can detect an effect of about 0.36 standard deviations (Section 5.7). Suppose the estimated effect is 0.20 standard deviations with a 95% confidence interval from −0.05 to 0.45. The honest statement is that the estimate is positive, that it is consistent with effects from slightly negative to large, and that the cohort is too small to settle the question. The provider would report this, pool it with the next cohort run under the same design, and say what it will do next. That is a rung C design with a precision problem. A provider that had instead reported the before-and-after change in the treated group, say 0.9 standard deviations, would have reported a number that mixes the programme's effect with maturation, practice effects on the test and selection, and that could mislead a funder by a factor of several. The illustration shows both the value of a comparison group and the importance of reporting uncertainty.

10.6 Governance of evidence inside a small organisation

Standards are kept by structures. A small provider can adopt three inexpensive governance practices. First, it can designate an evidence lead who is responsible for the logic models, instruments and reporting standard and who has the authority to report findings that the commercial or delivery team would prefer to suppress. Second, it can adopt a written publication policy that commits the organisation to publish evaluation results, including null and negative ones, within a stated time of data collection, and to share the underlying data in de-identified form where consent allows. Third, it can establish an advisory or evidence committee with at least one external member who has research expertise and no financial interest in the provider's results. These practices cost little. Their effect is to move the question of whether to report an unfavourable result from a decision made in the moment, under commercial pressure, to a decision already made in advance.

10.7 Evidence, equity and the learner who is hardest to reach

Evidence standards interact with equity in ways that are easily missed. Average effects can conceal large differences between groups. Muralidharan et al. (2019) found that gains were proportionally much larger for academically weaker students, and Kizilcec et al. (2017) found that brief interventions could narrow achievement gaps between learners in developed and less-developed countries. In both cases the informative result was in the disaggregation. A provider that reports only an average cannot say whether it is closing or widening gaps, and a funder interested in equity needs to know. The reporting standard therefore requires disaggregation by sex, disability status, age and location, which links the evidence principle to the first paper's access framework. Disaggregation has a cost: subgroups are small, and estimates are imprecise (Section 5.7). The remedy is to pre-specify the subgroups that matter, to pool across cohorts, and to treat imprecise subgroup results as signals for further study and not as conclusions.

11. Limitations

The paper has limitations that bear on its conclusions. It is a narrative review without a registered protocol and reports no screening counts, so selection bias cannot be excluded despite the steps in Section 2. Much of the experimental evidence in Section 4 comes from India and other regions and from school-age populations; the Kenyan studies concern early-grade literacy and school-based reforms, and the transfer of the findings to adult and workforce learning is plausible but unproven. The ESSA tiers originate in United States law and are used here as a convenient graduated scale, not as a local standard. The evidence ladder has not itself been validated, and the proposed minimum reporting standard has not been piloted with providers or funders. For several sources only an abstract or catalogue record could be accessed, and statements are confined to what those records report. Finally, the reflexive application is a self-assessment by an interested party. The propositions in Table 3 and the research agenda in Section 10.4 are offered so that these limits can be addressed with evidence.

12. Conclusion

Education technology and training will be judged, eventually, by what they change. The evidence suggests that what changes learning is not novelty but the match between instruction and learner, and the sustained support of the person who teaches. A small provider cannot always run a randomised trial, but it can always say what it measured, how, and what it does not know. A proportionate evidence standard turns that honesty into practice: claims no larger than the design supports, a public commitment to the next rung, independent analysis, and the reporting of costs and failures. The evidence ladder, the twelve-item reporting standard and the six propositions offered here are intended to make such practice routine, so

that funders can find providers that measure what changes, and providers can show that they do.

Declarations

Declaration of competing interest

The first author is the founder of Instracta eLearning Solutions, whose programmes are discussed in Section 9. No other competing interests are declared.

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Declaration of generative AI use

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Author contributions (CRediT)

Opiyo Michael Ouru: Conceptualization, Methodology, Investigation, Writing: original draft, Writing: review and editing.

Kenneth Goga Riany: Writing: review and editing.

Daisy Chepkoech O'maera: Writing: review and editing.

Data availability

No new data were generated. All sources are listed in the reference list; the search strings used are given in Appendix A.

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Appendix A. Search strings and source types

Searches were conducted in October 2026. Searches were iterative and results were screened by title and abstract against the criteria in Section 2.2.

Table A1. Principal search strings and source types

Topic	Search string (abridged)	Source types
Experimental evidence in Africa and India	Kenya randomised evaluation reading PRIMR; teacher professional development meta-analysis Sub-Saharan Africa; tracking Kenya randomised; remedying education India; technology-aided instruction India	AEA journals, NBER, World Development, Review of Educational Research, repository records
Technology in education	upgrading education with technology experimental research; low-tech learning school closures; mobile phones adult education Niger; technology teacher professional development LMIC systematic review	Journal of Economic Literature, Nature Human Behaviour, AEJ: Applied, Computers and Education Open
Training evaluation	Kirkpatrick four levels critique; training criteria meta-analysis; trainee reactions meta-analysis; transfer of training meta-analysis; training effectiveness meta-analysis	Personnel Psychology, Journal of Applied Psychology, Journal of Management, repository records
Evidence grading and validity	ESSA tiers of evidence; learning from experiments when context matters; how much can we generalize impact evaluations; publication bias file drawer; open research culture	CADRE, AER P&P, JEEA, Science
Policy and data	Global Education Monitoring Report 2023; Kenya Data Protection Act 2019	UNESCO, Republic of Kenya

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